Paper Reproduction: SimpleShot Revisiting Nearest-Neighbor Classification for Few-Shot Learning

Woody Chang, Jiarui Zhang, Qiao Xiao, Xinyu Tan

Cornell University

Abstract

Few-shot image classification addresses the problem of recognizing classes for which only a handful of labeled examples are available. This setting is important because many real-world tasks do not provide enough annotated data to support conventional deep learning pipelines. Traditional supervised learning methods typically require large-scale labeled datasets and tend to generalize poorly when only one or a few examples per class are available. A large body of prior work in few-shot learning has focused on meta-learning, episodic training [2, 3], or specially designed optimization procedures. In contrast, the SimpleShot paper [1] proposed a much simpler perspective: **strong few-shot performance** can emerge from high-quality supervised feature learning combined with a **simple nearest-class-mean** decision rule. This result is important because it challenges the assumption that increasingly complex few-shot methods are always necessary. The goal of this project is to reproduce the main empirical findings of SimpleShot on miniImageNet. Specifically, we investigate whether the relative performance trends reported in the original work can be recovered when using multiple standard backbones and feature normalization strategies. We also compare the performance of the final checkpoint at **epoch 90** with the **best saved checkpoint** in order to evaluate training stability and sensitivity to model selection.

Questions that we want to explore

1. Can the SimpleShot framework be successfully reproduced on miniImageNet using standard backbone networks?
2. Do deeper networks consistently yield better few-shot classification performance?
3. How much do feature normalization strategies such as L2N and CL2N improve performance over unnormalized features?
4. How different are the final epoch results from the best saved checkpoint results?
5. If our reproduced results fall below the original paper, what likely causes explain the gap?

Experiment Setup

- We conduct all experiments on **miniImageNet**, a standard benchmark for few-shot image classification. The dataset is split into base, validation, and test categories. The backbone is trained on the base split, while few-shot episodes are evaluated on novel classes from the test split.
- We follow the **5-way few-shot** training and testing scheme and for each configuration, **we evaluate the model over 10000 episodes and report the classification accuracy with confidence intervals**.
- The backbone feature extractor models we used are Conv4, ResNet10, ResNet18 [4], and DenseNet121 [5].
- Following the original paper [1], all models were trained for 90 epochs. To better analyze performance differences across models, we evaluated them under two settings. First, we ran the evaluation pipeline on each model after completing all 90 epochs. Second, we repeated the evaluation using the checkpoint corresponding to the epoch that achieved the highest validation accuracy for each model.

Parameters Selection

Following the official SimpleShot configuration [1], we adopted the core hyperparameters including an **SGD optimizer** (momentum 0.9), an **initial learning rate of 0.1** with a MultiStepLR schedule (milestones at [45, 66], $\gamma = 0.1$), and a **batch size of 256**. All models were trained for 90 epochs using cross-entropy loss on 64 base classes.

Modifications & Refinements: Since the original paper omitted specific details, we implemented these strategies inspired by canonical papers:

- **Standardized Input:** Applied ImageNet normalization ($\mu = [.485, .456, .406], \sigma = [.229, .224, .225]$). This ensures early-epoch loss decreases faster and final features are more discriminative.
- **Data Augmentation:** Used *RandomResizedCrop(84)* and *HorizontalFlip* to learn scale and pose invariance, preventing overfitting on the training split.
- **Weight Decay (1e-4):** Added L2 penalty to SGD to prevent feature geometry elongation. Since SimpleShot relies on Euclidean distance, which improves geometric quality.

Main Results

Our reproduction successfully recovers the main qualitative trends of the original SimpleShot paper [1]. Across both 1-shot and 5-shot evaluation, **deeper backbones outperform shallower ones**, and **normalized feature representations consistently improve performance over unnormalized features**. In particular, **CL2N usually achieves the highest accuracy** among the three transformation settings, confirming that both centering and normalization are beneficial for few-shot prototype-based classification.

At the same time, our reproduced results are generally lower than the values reported in the original paper [1]. The gap is typically around 2–3 percentage points for the stronger backbones, which is consistent with known sensitivity of reproduction results to **software versions and data preprocessing pipelines**.

SimpleShot Replication: All 4 Models vs Paper Table 1 (miniImageNet, 5-way, 10K episodes)

Model	Transform	1-shot Ours	1-shot Paper	Gap	5-shot Ours	5-shot Paper	Gap
conv4	UN	34.28+/-0.17	33.17+/-0.17	+1.11	62.35+/-0.17	63.25+/-0.17	-0.90
conv4	L2N	46.50+/-0.18	48.08+/-0.18	-1.58	64.48+/-0.17	66.49+/-0.17	-2.01
conv4	CL2N	48.03+/-0.19	49.69+/-0.19	-1.66	65.20+/-0.17	66.92+/-0.17	-1.72
resnet10	UN	51.86+/-0.21	54.45+/-0.21	-2.59	74.46+/-0.16	76.98+/-0.15	-2.52
resnet10	L2N	55.68+/-0.20	57.85+/-0.20	-2.17	75.81+/-0.16	78.73+/-0.15	-2.92
resnet10	CL2N	58.39+/-0.20	60.85+/-0.20	-2.46	75.80+/-0.16	78.40+/-0.15	-2.60
resnet18	UN	53.82+/-0.21	56.06+/-0.20	-2.24	76.56+/-0.15	78.63+/-0.15	-2.07
resnet18	L2N	57.72+/-0.20	60.16+/-0.20	-2.44	77.70+/-0.15	79.94+/-0.14	-2.24
resnet18	CL2N	60.30+/-0.20	62.85+/-0.20	-2.55	77.83+/-0.15	80.02+/-0.14	-2.19
densenet121	UN	55.54+/-0.20	57.81+/-0.21	-2.27	77.95+/-0.15	80.43+/-0.15	-2.48
densenet121	L2N	59.55+/-0.20	61.49+/-0.20	-1.94	78.60+/-0.15	81.48+/-0.14	-2.88
densenet121	CL2N	61.78+/-0.20	64.29+/-0.20	-2.51	79.10+/-0.15	81.50+/-0.14	-2.40

Figure 1. Performance evaluated based on epoch 90 for each model

SimpleShot Replication: All 4 Models vs Paper Table 1 (miniImageNet, 5-way, 10K episodes)

Model	Transform	1-shot Ours	1-shot Paper	Gap	5-shot Ours	5-shot Paper	Gap
conv4	UN	34.11+/-0.17	33.17+/-0.17	+0.94	62.19+/-0.17	63.25+/-0.17	-1.06
conv4	L2N	46.34+/-0.18	48.08+/-0.18	-1.74	64.52+/-0.17	66.49+/-0.17	-1.97
conv4	CL2N	47.98+/-0.19	49.69+/-0.19	-1.71	65.12+/-0.17	66.92+/-0.17	-1.80
resnet10	UN	52.01+/-0.21	54.45+/-0.21	-2.44	74.39+/-0.16	76.98+/-0.15	-2.59
resnet10	L2N	55.76+/-0.20	57.85+/-0.20	-2.09	75.77+/-0.15	78.73+/-0.15	-2.96
resnet10	CL2N	58.33+/-0.20	60.85+/-0.20	-2.52	75.75+/-0.16	78.40+/-0.15	-2.65
resnet18	UN	53.90+/-0.21	56.06+/-0.20	-2.16	76.47+/-0.15	78.63+/-0.15	-2.16
resnet18	L2N	58.05+/-0.20	60.16+/-0.20	-2.11	77.59+/-0.15	79.94+/-0.14	-2.35
resnet18	CL2N	60.34+/-0.20	62.85+/-0.20	-2.51	77.61+/-0.15	80.02+/-0.14	-2.41
densenet121	UN	55.40+/-0.20	57.81+/-0.21	-2.41	78.17+/-0.15	80.43+/-0.15	-2.26
densenet121	L2N	59.64+/-0.20	61.49+/-0.20	-1.85	78.81+/-0.14	81.48+/-0.14	-2.67
densenet121	CL2N	61.94+/-0.20	64.29+/-0.20	-2.35	79.08+/-0.15	81.50+/-0.14	-2.42

Figure 2. Performance evaluated based on the best epoch for each model

Observation

Our reproduction reveals **several consistent patterns** across all backbone architectures and evaluation settings. First, **performance improves steadily with model depth**, with Conv4 performing the worst and DenseNet121 achieving the best results. This suggests that stronger **feature representations directly translate to better few-shot generalization**.

Second, **feature normalization plays a crucial role**. Across all models, L2N consistently outperforms unnormalized features, while CL2N further improves performance in most cases. This indicates that **both scaling and centering of feature space significantly enhance the effectiveness of nearest-prototype classification**.

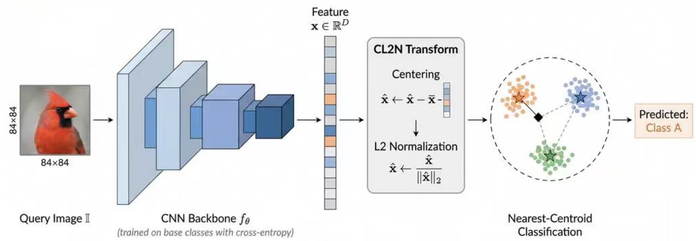
Third, **the difference between the final checkpoint at epoch 90 and the best saved checkpoint is minimal across all configurations**. This suggests that the **training process is stable** and that the models do not exhibit significant overfitting near the end of training.

Finally, although our results successfully reproduce the overall trends reported in the original paper, the absolute performance is consistently lower by approximately 2–3% for most settings. This gap is likely due to differences in implementation details, including the PyTorch version and dataset preprocessing pipeline.

Future work

Future work could improve reproduction fidelity by matching the original paper's software environment more closely, including the use of PyTorch 1.0 and the original raw image pipeline. It would also be useful to test **whether the same conclusions hold on other few-shot benchmarks such as CIFAR-FS, tieredImageNet**, or CUB. In addition, one could study how newer backbone families, including Vision Transformers or modern convolutional architectures, behave under the same SimpleShot evaluation framework. Finally, further work could analyze how centering and normalization influence class separation in feature space from a geometric perspective.

Overall Pipeline



No meta-learning required — simple feature transformations suffice for competitive few-shot learning.

Figure 3. Overall pipeline of the SimpleShot framework.

References

1. Y. Wang, W.-L. Chao, K. Q. Weinberger, and L. van der Maaten, "SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning," *arXiv preprint arXiv:1911.04623*, 2019.
2. O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching Networks for One Shot Learning," in *NeurIPS*, 2016.
3. J. Snell, K. Swersky, and R. Zemel, "Prototypical Networks for Few-shot Learning," in *NeurIPS*, 2017.
4. K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016.
5. G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *CVPR*, 2017.